Epcoritamab_mPBPK_QSP Implementation

Mechanistic Simulation of Bispecific Antibody Therapy in B-Cell Lymphoma

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Contents

1. Introduction

This report implements the semi-mechanistic PBPK/PD model for Epcoritamab (CD3xCD20), based on Li et al. (2022). The model integrates five biological sub-systems to predict tumor response from first principles:
1. **Pharmacokinetics (PK):** Two-pore distribution of the antibody.
2. **Trafficking:** Recirculation of T-cells and B-cells between blood and lymphoid tissues.
3. **Binding:** Kinetic formation of the pharmacologically active **Trimer** ($T - Drug - B$).
4. **Activation:** Trimer-dependent activation and expansion of cytotoxic T-cells.
5. **Killing:** Elimination of B-cells and Tumor cells by activated T-cells.

2. Model Structure

The model is implemented in C++ using `mrgsolve`. It integrates five submodels (PK, Trafficking, Binding, Activation, Killing) into a single system of differential equations.

```
model_code <- '
$PARAM @annotated
// --- PK Parameters ---
ka          : 0.131   : Absorption rate (1/day)
CL          : 2.47    : Clearance (L/day)
Km          : 0.0461  : MM constant (nM)
Vmax        : 0.185   : Max nonlinear clearance (nM/day)
Vplasma     : 2.6     : Plasma volume (L)
Vleaky      : 4.37    : Leaky tissue volume (L)
Vtight      : 8.11    : Tight tissue volume (L)
Vspleen     : 0.0433  : Spleen volume (L)
Vnode       : 0.0082  : Node volume (L)
Vlymph      : 5.2     : Lymph volume (L)
Lleaky      : 1.94    : Flow leaky (L/day)
```

```
Ltight      : 0.957   : Flow tight (L/day)
Lspleen     : 0.304   : Flow spleen (L/day)
L           : 2.9     : Total lymph flow (L/day)
sigma_leaky : 0.85    : Sigma leaky
sigma_tight : 0.95    : Sigma tight
sigma_spleen: 0.85    : Sigma spleen
sigma_lymph : 0.2     : Sigma lymph

// --- Trafficking Parameters ---
kinTC       : 310e6   : TC production (cells/day)
koutTC      : 0.00422 : TC death (1/day)
kinBC       : 223e6   : BC production (cells/day)
koutBC      : 0.0189  : BC death (1/day)
kpt         : 50      : Blood to Spleen (1/day)
ktn         : 1.67    : Spleen to Node (1/day)
knl         : 1672    : Node to Lymph (1/day)
klp         : 2.63    : Lymph to Blood (1/day)
kdecay      : 0.681   : Injection effect decay (1/day)
INJ_Scaler  : 9.08    : Injection effect magnitude
kt          : 0.00027 : Homeostasis rate (1/day)
r           : 2.24    : Homeostasis strength
TC_base_count : 7.345e9 : Baseline Total T-Cells in Blood
BC_base_count : 1.18e9  : Baseline Total B-Cells in Blood

// --- Binding Parameters ---
konCD3      : 18.1    : Assoc CD3 (L/nmol/day)
koffCD3     : 285     : Dissoc CD3 (1/day)
konCD20     : 4.15    : Assoc CD20 (L/nmol/day)
koffCD20    : 22.5    : Dissoc CD20 (1/day)
kdegCD3     : 1.584   : Degradation CD3 (1/day)
kdegCD20    : 1.584   : Degradation CD20 (1/day)
kintCD3     : 1.584   : Internalization CD3 (1/day)
kintCD20    : 1.584   : Internalization CD20 (1/day)
R_CD3       : 30000   : Receptors/TC
R_CD20      : 100000  : Receptors/BC
scale_binding : 1.0     : Stiffness control

// --- Activation & Killing Parameters ---
sim_slope        : 0.007e6 : Activation rate vs BC
sim_slopetumor   : 7  : Activation rate vs Tumor
expand_factor    : 9.27  : T-cell expansion (1/day)
koutATC          : 0.05  : ATC death (1/day)
TAD              : 3.0   : Activation delay (days)
Tp               : 10.8  : Proliferation duration (days)
Trimer_Threshold : 24    : Threshold for tumor activation
kkill_BC         : 0.544e-6 : Killing rate constant (L^2/cells/day)
Time_First_Dose  : 0.0        : Time when the first dose is administered

// --- Tumor Parameters ---
kgrowth_DLBCL    : 0.0301     : Tumor growth rate (1/day)
Tumor_r0         : 1.40       : Initial tumor radius (cm)
tumor_capacity   : 1.0e-12    : Capacity factor (1/cell)
depth            : 0.01       : Reachable depth (cm)
kkill_tumor      : 0.165e-6   : Killing rate Tumor (L^2/cells/day)
```

```
R_CD20_tumor     : 30000      : Receptors/Tumor Cell

$CMT
// PK Compartments
DEPOT PLASMA LEAKY TIGHT SPLEEN NODE LYMPH

// Lymphocyte Compartments (TC = T-cells, BC = B-cells)
TC_BLOOD TC_SPLEEN TC_NODE TC_LYMPH
BC_BLOOD BC_SPLEEN BC_NODE BC_LYMPH
INJ
AF_TC //adaptive feedback
AF_BC

// Binding Compartments (All tissues healthy)
FREE_CD3_BLOOD  FREE_CD20_BLOOD  DIMER_CD3_BLOOD  DIMER_CD20_BLOOD  TRIMER_BLOOD
FREE_CD3_SPLEEN FREE_CD20_SPLEEN DIMER_CD3_SPLEEN DIMER_CD20_SPLEEN TRIMER_SPLEEN
FREE_CD3_NODE   FREE_CD20_NODE   DIMER_CD3_NODE   DIMER_CD20_NODE   TRIMER_NODE
FREE_CD3_LYMPH  FREE_CD20_LYMPH  DIMER_CD3_LYMPH  DIMER_CD20_LYMPH  TRIMER_LYMPH

// ACTIVATION
vATC_BC_BLOOD vATC_BC_SPLEEN vATC_BC_NODE vATC_BC_LYMPH
vATC_TUMOR_NODE
pATC_BLOOD pATC_SPLEEN pATC_NODE pATC_LYMPH
// TUMOR
TUMOR_CELLS
FREE_CD20_TUMOR DIMER_CD20_TUMOR TRIMER_TUMOR

// CLOCK
T_SIM

$GLOBAL
#define CLAMP(x) ((x) > 0 ? (x) : 0.0)
// Constants
double AVOG = 6.022e23;
double nmol_per_molecule = 1e9 / AVOG; // Helper to convert counts to nmol
double PI = 3.14159265359; //Helper to calculate volume

$MAIN
// Auto-Initialization
if(NEWIND <= 1) {
  // Binding: Init based on baseline cells
  FREE_CD3_BLOOD_0  = TC_BLOOD_0 * R_CD3  * nmol_per_molecule;
  FREE_CD20_BLOOD_0 = BC_BLOOD_0 * R_CD20 * nmol_per_molecule;
  FREE_CD3_SPLEEN_0 = TC_SPLEEN_0 * R_CD3  * nmol_per_molecule;
  FREE_CD20_SPLEEN_0= BC_SPLEEN_0 * R_CD20 * nmol_per_molecule;
  FREE_CD3_NODE_0   = TC_NODE_0 * R_CD3  * nmol_per_molecule;
  FREE_CD20_NODE_0  = BC_NODE_0 * R_CD20 * nmol_per_molecule;
  FREE_CD3_LYMPH_0  = TC_LYMPH_0 * R_CD3  * nmol_per_molecule;
  FREE_CD20_LYMPH_0 = BC_LYMPH_0 * R_CD20 * nmol_per_molecule;
  // 2. Tumor Init
  double Vol_Tumor_mL_0 = (4.0/3.0) * PI * pow(Tumor_r0, 3.0);
  TUMOR_CELLS_0 = Vol_Tumor_mL_0 * 1e9;
  // Initialize Tumor Binding
  FREE_CD20_TUMOR_0 = TUMOR_CELLS_0 * R_CD20_tumor * nmol_per_molecule; //unit: nmol
```

```
  // Initialize Clock
  T_SIM_0 = 0;
}


$ODE
{
  // --- CLOCK ODE ---
  dxdt_T_SIM = 1;

  // ------------------------------------------------------------------
  // 1. PK SUBMODEL EQUATIONS
  // ------------------------------------------------------------------
  double C_plasma = PLASMA / Vplasma; //nmol/L
  double C_leaky  = LEAKY  / Vleaky;
  double C_tight  = TIGHT  / Vtight;
  double C_spleen = SPLEEN / Vspleen;
  double C_node   = NODE   / Vnode;
  double C_lymph  = LYMPH  / Vlymph;

  double dist_plasma_leaky  = Lleaky  * (1.0 - sigma_leaky)  * C_plasma;
  double dist_plasma_tight  = Ltight  * (1.0 - sigma_tight)  * C_plasma;
  double dist_plasma_spleen = Lspleen * (1.0 - sigma_spleen) * C_plasma;
  double dist_leaky_lymph   = Lleaky  * (1.0 - sigma_lymph)  * C_leaky;
  double dist_tight_lymph   = Ltight  * (1.0 - sigma_lymph)  * C_tight;
  double dist_spleen_node   = Lspleen * (1.0 - sigma_lymph)  * C_spleen;
  double dist_node_lymph    = Lspleen * C_node;
  double dist_lymph_plasma  = L * C_lymph;

  double elim_linear    = CL * C_plasma;
  double elim_nonlinear = (Vmax * C_plasma) / (Km + C_plasma) * Vplasma;

  dxdt_DEPOT  = -ka * DEPOT;
  dxdt_PLASMA = -elim_linear - elim_nonlinear - dist_plasma_leaky - dist_plasma_tight -
↪  dist_plasma_spleen + dist_lymph_plasma;
  dxdt_LEAKY  = dist_plasma_leaky - dist_leaky_lymph;
  dxdt_TIGHT  = dist_plasma_tight - dist_tight_lymph;
  dxdt_SPLEEN = dist_plasma_spleen - dist_spleen_node;
  dxdt_NODE   = dist_spleen_node - dist_node_lymph;
  dxdt_LYMPH  = (ka * DEPOT) + dist_leaky_lymph + dist_tight_lymph + dist_node_lymph -
↪  dist_lymph_plasma;

  // ------------------------------------------------------------------
  // 2. LYMPHOCYTE TRAFFICKING SUBMODEL
  // ------------------------------------------------------------------
  // The "Injection Effect": Drives cells out of blood into spleen temporarily
  dxdt_INJ = -kdecay * INJ;

  double TC_Current_Total = TC_BLOOD + pATC_BLOOD;
  double Drive_TC = TC_base_count / (TC_Current_Total > 1.0 ? TC_Current_Total : 1.0);
  dxdt_AF_TC = kt * (Drive_TC - AF_TC);

  double BC_Current_Total = BC_BLOOD;
  double Drive_BC = BC_base_count / (BC_Current_Total > 1.0 ? BC_Current_Total : 1.0);
```

```
dxdt_AF_BC = kt * (Drive_BC - AF_BC);

double prod_TC = kinTC * pow(CLAMP(AF_TC), r);
double prod_BC = kinBC * pow(CLAMP(AF_BC), r);

// Rate constants for flows //Unit: day^(-1)
double rate_BL_SP = kpt * (1 + INJ_Scaler * INJ);
double rate_SP_LN = ktn;
double rate_LN_LY = knl;
double rate_LY_BL = klp;

// ============================================================
// 3. ACTIVATION VARIABLES //Unit: Cells/L
// ============================================================
double C_vATC_BL = vATC_BC_BLOOD / Vplasma;
double C_vATC_SP = vATC_BC_SPLEEN / Vspleen;
double C_vATC_LN = vATC_BC_NODE   / Vnode;
double C_vATC_LY = vATC_BC_LYMPH  / Vlymph;
double C_vATC_Tumor = vATC_TUMOR_NODE / Vnode;

// ============================================================
// 4. B-CELL KILLING SUBMODEL
// ============================================================
double C_BC_BL   = BC_BLOOD / Vplasma;
double C_BC_SP   = BC_SPLEEN / Vspleen;
double C_BC_LN   = BC_NODE   / Vnode;
double C_BC_LY   = BC_LYMPH  / Vlymph;

// Calculate Killing RATES (Cells/Day)
// Formula: kkill * C_vATC * C_BC
// the inputs to the killing function should be concentrations (Cells/L).
double rate_kill_BC_BL = kkill_BC * C_vATC_BL * C_BC_BL;
double rate_kill_BC_SP = kkill_BC * C_vATC_SP * C_BC_SP;
double rate_kill_BC_LN = kkill_BC * C_vATC_LN * C_BC_LN;
double rate_kill_BC_LY = kkill_BC * C_vATC_LY * C_BC_LY;

// Effective Death Rate (1/Day)
double k_kill_BL = (BC_BLOOD > 1.0)  ? (rate_kill_BC_BL / BC_BLOOD)  : 0.0;
double k_kill_SP = (BC_SPLEEN > 1.0) ? (rate_kill_BC_SP / BC_SPLEEN) : 0.0;
double k_kill_LN = (BC_NODE > 1.0)   ? (rate_kill_BC_LN / BC_NODE)   : 0.0;
double k_kill_LY = (BC_LYMPH > 1.0)  ? (rate_kill_BC_LY / BC_LYMPH)  : 0.0;

// ============================================================
// 5. TUMOR SUBMODEL
// ============================================================
double Cells_Total = CLAMP(TUMOR_CELLS); //Total number of tumor cells in the tumor
↪  lymph node

// A. Geometry
// This assumes 1 mL of tumor contain 1e9 tumor cells.
double Vol_Tumor_mL = Cells_Total / 1e9;
double R_tumor = pow( (3.0 * Vol_Tumor_mL) / (4.0 * PI), 1.0/3.0 );

double Vol_Reachable_mL = 0.0;
```

```
  if (R_tumor > depth) {
    double R_core = R_tumor - depth;
    Vol_Reachable_mL = Vol_Tumor_mL - ((4.0/3.0) * PI * pow(R_core, 3.0));
  } else {
    Vol_Reachable_mL = Vol_Tumor_mL;
  }
  double Cells_Reachable = Vol_Reachable_mL * 1e9;
  double C_Tumor_Reachable = Cells_Reachable / Vnode;
  // B. Tumor Growth //unit: cell/day
  double rate_growth = kgrowth_DLBCL * Cells_Total * (1.0 - tumor_capacity *
↪  Cells_Total);
  // C. Tumor Killing
  double rate_kill_Tumor = kkill_tumor * C_vATC_Tumor * C_Tumor_Reachable; //unit:
↪  L^2/cells/day * cell/L * cell/L = cell/day
  double k_kill_Tumor_frac = (Cells_Reachable > 1.0) ? (rate_kill_Tumor /
↪  Cells_Reachable) : 0.0; // unit: 1/day

  dxdt_TUMOR_CELLS = rate_growth - rate_kill_Tumor;

  // ============================================================
  // 6. TRAFFICKING ODES Unit: cells/day
  // ============================================================
  // T-Cells (No killing, just flow & death)
  dxdt_TC_BLOOD  = prod_TC - koutTC * TC_BLOOD - rate_BL_SP * TC_BLOOD + rate_LY_BL *
↪  TC_LYMPH;
  dxdt_TC_SPLEEN = rate_BL_SP * TC_BLOOD - rate_SP_LN * TC_SPLEEN - koutTC * TC_SPLEEN;
  dxdt_TC_NODE   = rate_SP_LN * TC_SPLEEN - rate_LN_LY * TC_NODE - koutTC * TC_NODE;
  dxdt_TC_LYMPH  = rate_LN_LY * TC_NODE - rate_LY_BL * TC_LYMPH - koutTC * TC_LYMPH;

  // B-Cells (Trafficking Flow + Death + KILLING)
  dxdt_BC_BLOOD  = prod_BC - koutBC * BC_BLOOD - rate_BL_SP * BC_BLOOD + rate_LY_BL *
↪  BC_LYMPH - rate_kill_BC_BL;
  dxdt_BC_SPLEEN = rate_BL_SP * BC_BLOOD - rate_SP_LN * BC_SPLEEN - koutBC * BC_SPLEEN -
↪  rate_kill_BC_SP;
  dxdt_BC_NODE   = rate_SP_LN * BC_SPLEEN - rate_LN_LY * BC_NODE - koutBC * BC_NODE -
↪  rate_kill_BC_LN;
  dxdt_BC_LYMPH  = rate_LN_LY * BC_NODE - rate_LY_BL * BC_LYMPH - koutBC * BC_LYMPH -
↪  rate_kill_BC_LY;

  // ============================================================
  // 7. BINDING SUBMODEL
  // ============================================================
  // Helper variables for binding concentrations //unit: nmol/L
  double C_FreeCD3_BL   = FREE_CD3_BLOOD / Vplasma; //FREE_CD3_BLOOd is in nmol
  double C_FreeCD20_BL  = FREE_CD20_BLOOD / Vplasma;
  double C_DimerCD3_BL  = DIMER_CD3_BLOOD / Vplasma;
  double C_DimerCD20_BL = DIMER_CD20_BLOOD / Vplasma;
  double C_Trimer_BL    = TRIMER_BLOOD / Vplasma;

  double C_FreeCD3_SP = FREE_CD3_SPLEEN/Vspleen; double C_FreeCD20_SP =
↪  FREE_CD20_SPLEEN/Vspleen;
  double C_DimerCD3_SP = DIMER_CD3_SPLEEN/Vspleen; double C_DimerCD20_SP =
↪  DIMER_CD20_SPLEEN/Vspleen;
  double C_Trimer_SP = TRIMER_SPLEEN/Vspleen;
```

```
double C_FreeCD3_LN = FREE_CD3_NODE/Vnode; double C_FreeCD20_LN = FREE_CD20_NODE/Vnode;
double C_DimerCD3_LN = DIMER_CD3_NODE/Vnode; double C_DimerCD20_LN =
↪  DIMER_CD20_NODE/Vnode;
double C_Trimer_LN = TRIMER_NODE/Vnode;

double C_FreeCD3_LY = FREE_CD3_LYMPH/Vlymph; double C_FreeCD20_LY =
↪  FREE_CD20_LYMPH/Vlymph;
double C_DimerCD3_LY = DIMER_CD3_LYMPH/Vlymph; double C_DimerCD20_LY =
↪  DIMER_CD20_LYMPH/Vlymph;
double C_Trimer_LY = TRIMER_LYMPH/Vlymph;

// Tumor Binding Vars (in node only)
double C_FreeCD20_Tumor = FREE_CD20_TUMOR / Vnode;
double C_DimerCD20_Tumor = DIMER_CD20_TUMOR / Vnode;
double C_Trimer_Tumor    = TRIMER_TUMOR / Vnode;

// --- Velocities //unit: nmol/day ---
// BLOOD
double vf_DimerCD3_BL  = scale_binding * konCD3 * C_plasma * C_FreeCD3_BL * Vplasma;
double vf_DimerCD20_BL = scale_binding * konCD20 * C_plasma * C_FreeCD20_BL * Vplasma;
double vb_DimerCD3_BL  = scale_binding * koffCD3 * C_DimerCD3_BL * Vplasma;
double vb_DimerCD20_BL = scale_binding * koffCD20 * C_DimerCD20_BL * Vplasma;
double vf_Tri_viaCD3_BL  = scale_binding * konCD20 * C_DimerCD3_BL * C_FreeCD20_BL *
↪  Vplasma;
double vf_Tri_viaCD20_BL = scale_binding * konCD3 * C_DimerCD20_BL * C_FreeCD3_BL *
↪  Vplasma;
double vb_Tri_toCD3_BL   = scale_binding * koffCD20 * C_Trimer_BL * Vplasma;
double vb_Tri_toCD20_BL  = scale_binding * koffCD3 * C_Trimer_BL * Vplasma;

// SPLEEN
double vf_DimerCD3_SP  = scale_binding * konCD3 * C_spleen * C_FreeCD3_SP * Vspleen;
double vf_DimerCD20_SP = scale_binding * konCD20 * C_spleen * C_FreeCD20_SP * Vspleen;
double vb_DimerCD3_SP  = scale_binding * koffCD3 * C_DimerCD3_SP * Vspleen;
double vb_DimerCD20_SP = scale_binding * koffCD20 * C_DimerCD20_SP * Vspleen;
double vf_Tri_viaCD3_SP  = scale_binding * konCD20 * C_DimerCD3_SP * C_FreeCD20_SP *
↪  Vspleen;
double vf_Tri_viaCD20_SP = scale_binding * konCD3 * C_DimerCD20_SP * C_FreeCD3_SP *
↪  Vspleen;
double vb_Tri_toCD3_SP   = scale_binding * koffCD20 * C_Trimer_SP * Vspleen;
double vb_Tri_toCD20_SP  = scale_binding * koffCD3 * C_Trimer_SP * Vspleen;

// NODE (Healthy)
double vf_DimerCD3_LN  = scale_binding * konCD3 * C_node * C_FreeCD3_LN * Vnode;
double vf_DimerCD20_LN = scale_binding * konCD20 * C_node * C_FreeCD20_LN * Vnode;
double vb_DimerCD3_LN  = scale_binding * koffCD3 * C_DimerCD3_LN * Vnode;
double vb_DimerCD20_LN = scale_binding * koffCD20 * C_DimerCD20_LN * Vnode;
double vf_Tri_viaCD3_LN  = scale_binding * konCD20 * C_DimerCD3_LN * C_FreeCD20_LN *
↪  Vnode;
double vf_Tri_viaCD20_LN = scale_binding * konCD3 * C_DimerCD20_LN * C_FreeCD3_LN *
↪  Vnode;
double vb_Tri_toCD3_LN   = scale_binding * koffCD20 * C_Trimer_LN * Vnode;
double vb_Tri_toCD20_LN  = scale_binding * koffCD3 * C_Trimer_LN * Vnode;
```

```
// NODE (Tumor)
double vf_DimerCD20_Tumor = scale_binding * konCD20 * C_node * C_FreeCD20_Tumor *
↪  Vnode;
double vb_DimerCD20_Tumor = scale_binding * koffCD20 * C_DimerCD20_Tumor * Vnode;
double vf_Tri_Tumor_viaCD3 = scale_binding * konCD20 * C_DimerCD3_LN * C_FreeCD20_Tumor
↪  * Vnode;
double vf_Tri_Tumor_viaCD20= scale_binding * konCD3 * C_DimerCD20_Tumor * C_FreeCD3_LN
↪  * Vnode;
double vb_Tri_Tumor_toCD3  = scale_binding * koffCD20 * C_Trimer_Tumor * Vnode;
double vb_Tri_Tumor_toCD20 = scale_binding * koffCD3 * C_Trimer_Tumor * Vnode;

// LYMPH
double vf_DimerCD3_LY  = scale_binding * konCD3 * C_lymph * C_FreeCD3_LY * Vlymph;
double vf_DimerCD20_LY = scale_binding * konCD20 * C_lymph * C_FreeCD20_LY * Vlymph;
double vb_DimerCD3_LY  = scale_binding * koffCD3 * C_DimerCD3_LY * Vlymph;
double vb_DimerCD20_LY = scale_binding * koffCD20 * C_DimerCD20_LY * Vlymph;
double vf_Tri_viaCD3_LY  = scale_binding * konCD20 * C_DimerCD3_LY * C_FreeCD20_LY *
↪  Vlymph;
double vf_Tri_viaCD20_LY = scale_binding * konCD3 * C_DimerCD20_LY * C_FreeCD3_LY *
↪  Vlymph;
double vb_Tri_toCD3_LY   = scale_binding * koffCD20 * C_Trimer_LY * Vlymph;
double vb_Tri_toCD20_LY  = scale_binding * koffCD3 * C_Trimer_LY * Vlymph;

// --- Syntheses of Free CD3 and CD20 due to Natural Turnover //unit: nmol/day ---
double syn_CD3_BL  = kdegCD3 * CLAMP(TC_BLOOD) * R_CD3 * nmol_per_molecule;
double syn_CD20_BL = kdegCD20 * CLAMP(BC_BLOOD) * R_CD20 * nmol_per_molecule;
double syn_CD3_SP  = kdegCD3 * CLAMP(TC_SPLEEN) * R_CD3 * nmol_per_molecule;
double syn_CD20_SP = kdegCD20 * CLAMP(BC_SPLEEN) * R_CD20 * nmol_per_molecule;
double syn_CD3_LN  = kdegCD3 * CLAMP(TC_NODE) * R_CD3 * nmol_per_molecule;
double syn_CD20_LN = kdegCD20 * CLAMP(BC_NODE) * R_CD20 * nmol_per_molecule;
double syn_CD3_LY  = kdegCD3 * CLAMP(TC_LYMPH) * R_CD3 * nmol_per_molecule;
double syn_CD20_LY = kdegCD20 * CLAMP(BC_LYMPH) * R_CD20 * nmol_per_molecule;

// --- Flows ---
// Blood
double flow_FreeCD3_BL   = rate_LY_BL * FREE_CD3_LYMPH   - rate_BL_SP * FREE_CD3_BLOOD;
double flow_FreeCD20_BL  = rate_LY_BL * FREE_CD20_LYMPH - rate_BL_SP * FREE_CD20_BLOOD;
double flow_DimerCD3_BL  = rate_LY_BL * DIMER_CD3_LYMPH - rate_BL_SP * DIMER_CD3_BLOOD;
double flow_DimerCD20_BL = rate_LY_BL * DIMER_CD20_LYMPH- rate_BL_SP *
↪  DIMER_CD20_BLOOD;
double flow_Trimer_BL    = rate_LY_BL * TRIMER_LYMPH    - rate_BL_SP * TRIMER_BLOOD;
// Spleen
double flow_FreeCD3_SP   = rate_BL_SP * FREE_CD3_BLOOD  - rate_SP_LN * FREE_CD3_SPLEEN;
double flow_FreeCD20_SP  = rate_BL_SP * FREE_CD20_BLOOD - rate_SP_LN *
↪  FREE_CD20_SPLEEN;
double flow_DimerCD3_SP  = rate_BL_SP * DIMER_CD3_BLOOD - rate_SP_LN *
↪  DIMER_CD3_SPLEEN;
double flow_DimerCD20_SP = rate_BL_SP * DIMER_CD20_BLOOD- rate_SP_LN *
↪  DIMER_CD20_SPLEEN;
double flow_Trimer_SP    = rate_BL_SP * TRIMER_BLOOD    - rate_SP_LN * TRIMER_SPLEEN;
// Node
double flow_FreeCD3_LN   = rate_SP_LN * FREE_CD3_SPLEEN - rate_LN_LY * FREE_CD3_NODE;
double flow_FreeCD20_LN  = rate_SP_LN * FREE_CD20_SPLEEN- rate_LN_LY * FREE_CD20_NODE;
double flow_DimerCD3_LN  = rate_SP_LN * DIMER_CD3_SPLEEN- rate_LN_LY * DIMER_CD3_NODE;
```

```
double flow_DimerCD20_LN = rate_SP_LN * DIMER_CD20_SPLEEN-rate_LN_LY * DIMER_CD20_NODE;
double flow_Trimer_LN    = rate_SP_LN * TRIMER_SPLEEN    - rate_LN_LY * TRIMER_NODE;
// Lymph
double flow_FreeCD3_LY   = rate_LN_LY * FREE_CD3_NODE   - rate_LY_BL * FREE_CD3_LYMPH;
double flow_FreeCD20_LY  = rate_LN_LY * FREE_CD20_NODE  - rate_LY_BL * FREE_CD20_LYMPH;
double flow_DimerCD3_LY  = rate_LN_LY * DIMER_CD3_NODE  - rate_LY_BL * DIMER_CD3_LYMPH;
double flow_DimerCD20_LY = rate_LN_LY * DIMER_CD20_NODE - rate_LY_BL *
↪  DIMER_CD20_LYMPH;
double flow_Trimer_LY    = rate_LN_LY * TRIMER_NODE     - rate_LY_BL * TRIMER_LYMPH;

// --- Production ---
double prod_new_CD3_BL  = prod_TC * R_CD3 * nmol_per_molecule;
double prod_new_CD20_BL = prod_BC * R_CD20 * nmol_per_molecule;

// --- ODES: BINDING ---
// BLOOD
dxdt_FREE_CD3_BLOOD   = -vf_DimerCD3_BL + vb_DimerCD3_BL - vf_Tri_viaCD20_BL +
↪  vb_Tri_toCD20_BL
+ syn_CD3_BL - kdegCD3 * C_FreeCD3_BL * Vplasma + flow_FreeCD3_BL
+ prod_new_CD3_BL - koutTC * C_FreeCD3_BL * Vplasma;

dxdt_FREE_CD20_BLOOD  = -vf_DimerCD20_BL + vb_DimerCD20_BL - vf_Tri_viaCD3_BL +
↪  vb_Tri_toCD3_BL
+ syn_CD20_BL - kdegCD20 * C_FreeCD20_BL * Vplasma + flow_FreeCD20_BL
+ prod_new_CD20_BL - (koutBC + k_kill_BL) * C_FreeCD20_BL * Vplasma;

dxdt_DIMER_CD3_BLOOD  = vf_DimerCD3_BL - vb_DimerCD3_BL - vf_Tri_viaCD3_BL +
↪  vb_Tri_toCD3_BL
- kintCD3 * C_DimerCD3_BL * Vplasma - koutTC * C_DimerCD3_BL * Vplasma +
↪  flow_DimerCD3_BL
+ (koutBC + k_kill_BL) * C_Trimer_BL * Vplasma;

dxdt_DIMER_CD20_BLOOD = vf_DimerCD20_BL - vb_DimerCD20_BL - vf_Tri_viaCD20_BL +
↪  vb_Tri_toCD20_BL
- kintCD20 * C_DimerCD20_BL * Vplasma + flow_DimerCD20_BL
+ koutTC * C_Trimer_BL * Vplasma
- (koutBC + k_kill_BL) * C_DimerCD20_BL * Vplasma;

dxdt_TRIMER_BLOOD     = vf_Tri_viaCD3_BL + vf_Tri_viaCD20_BL - vb_Tri_toCD3_BL -
↪  vb_Tri_toCD20_BL
+ flow_Trimer_BL
- (koutTC + koutBC + k_kill_BL) * C_Trimer_BL * Vplasma;

// SPLEEN
dxdt_FREE_CD3_SPLEEN   = -vf_DimerCD3_SP + vb_DimerCD3_SP - vf_Tri_viaCD20_SP +
↪  vb_Tri_toCD20_SP - kdegCD3 * C_FreeCD3_SP * Vspleen + syn_CD3_SP + flow_FreeCD3_SP -
↪  koutTC * C_FreeCD3_SP * Vspleen;
dxdt_FREE_CD20_SPLEEN  = -vf_DimerCD20_SP + vb_DimerCD20_SP - vf_Tri_viaCD3_SP +
↪  vb_Tri_toCD3_SP - kdegCD20 * C_FreeCD20_SP * Vspleen + syn_CD20_SP + flow_FreeCD20_SP
↪  - (koutBC + k_kill_SP) * C_FreeCD20_SP * Vspleen;
dxdt_DIMER_CD3_SPLEEN  = vf_DimerCD3_SP - vb_DimerCD3_SP - vf_Tri_viaCD3_SP +
↪  vb_Tri_toCD3_SP - kintCD3 * C_DimerCD3_SP * Vspleen - koutTC * C_DimerCD3_SP *
↪  Vspleen + flow_DimerCD3_SP + (koutBC + k_kill_SP) * C_Trimer_SP * Vspleen;
dxdt_DIMER_CD20_SPLEEN = vf_DimerCD20_SP - vb_DimerCD20_SP - vf_Tri_viaCD20_SP +
↪  vb_Tri_toCD20_SP - kintCD20 * C_DimerCD20_SP * Vspleen + flow_DimerCD20_SP + koutTC *
↪  C_Trimer_SP * Vspleen - (koutBC + k_kill_SP) * C_DimerCD20_SP * Vspleen;
```

```
 dxdt_TRIMER_SPLEEN     = vf_Tri_viaCD3_SP + vf_Tri_viaCD20_SP - vb_Tri_toCD3_SP -
↪  vb_Tri_toCD20_SP + flow_Trimer_SP - (koutTC + koutBC + k_kill_SP) * C_Trimer_SP *
↪  Vspleen;

 // NODE
 dxdt_FREE_CD3_NODE   = -vf_DimerCD3_LN + vb_DimerCD3_LN - vf_Tri_viaCD20_LN +
↪  vb_Tri_toCD20_LN - kdegCD3 * C_FreeCD3_LN * Vnode + syn_CD3_LN + flow_FreeCD3_LN -
↪  koutTC * C_FreeCD3_LN * Vnode
 - vf_Tri_Tumor_viaCD20 + vb_Tri_Tumor_toCD20;

 dxdt_FREE_CD20_NODE  = -vf_DimerCD20_LN + vb_DimerCD20_LN - vf_Tri_viaCD3_LN +
↪  vb_Tri_toCD3_LN - kdegCD20 * C_FreeCD20_LN * Vnode + syn_CD20_LN + flow_FreeCD20_LN -
↪  (koutBC + k_kill_LN) * C_FreeCD20_LN * Vnode;

 dxdt_DIMER_CD3_NODE  = vf_DimerCD3_LN - vb_DimerCD3_LN - vf_Tri_viaCD3_LN +
↪  vb_Tri_toCD3_LN - kintCD3 * C_DimerCD3_LN * Vnode - koutTC * C_DimerCD3_LN * Vnode +
↪  flow_DimerCD3_LN
 + (koutBC + k_kill_LN) * C_Trimer_LN * Vnode
 - vf_Tri_Tumor_viaCD3 + vb_Tri_Tumor_toCD3
 + k_kill_Tumor_frac * TRIMER_TUMOR;

 dxdt_DIMER_CD20_NODE = vf_DimerCD20_LN - vb_DimerCD20_LN - vf_Tri_viaCD20_LN +
↪  vb_Tri_toCD20_LN - kintCD20 * C_DimerCD20_LN * Vnode + flow_DimerCD20_LN + koutTC *
↪  C_Trimer_LN * Vnode - (koutBC + k_kill_LN) * C_DimerCD20_LN * Vnode;

 dxdt_TRIMER_NODE     = vf_Tri_viaCD3_LN + vf_Tri_viaCD20_LN - vb_Tri_toCD3_LN -
↪  vb_Tri_toCD20_LN + flow_Trimer_LN - (koutTC + koutBC + k_kill_LN) * C_Trimer_LN *
↪  Vnode;

 // NODE (Tumor Binding)
 double syn_CD20_Tumor = kdegCD20 * Cells_Reachable * R_CD20_tumor * nmol_per_molecule;

 dxdt_FREE_CD20_TUMOR  = -vf_DimerCD20_Tumor + vb_DimerCD20_Tumor - vf_Tri_Tumor_viaCD3
↪  + vb_Tri_Tumor_toCD3
 - kdegCD20 * C_FreeCD20_Tumor * Vnode + syn_CD20_Tumor
 - k_kill_Tumor_frac * FREE_CD20_TUMOR;

 dxdt_DIMER_CD20_TUMOR = vf_DimerCD20_Tumor - vb_DimerCD20_Tumor - vf_Tri_Tumor_viaCD20
↪  + vb_Tri_Tumor_toCD20
 - kintCD20 * C_DimerCD20_Tumor * Vnode
 + koutTC * C_Trimer_Tumor * Vnode
 - k_kill_Tumor_frac * DIMER_CD20_TUMOR;

 dxdt_TRIMER_TUMOR     = vf_Tri_Tumor_viaCD3 + vf_Tri_Tumor_viaCD20 - vb_Tri_Tumor_toCD3
↪  - vb_Tri_Tumor_toCD20
 - koutTC * C_Trimer_Tumor * Vnode
 - k_kill_Tumor_frac * TRIMER_TUMOR;

 // LYMPH
 dxdt_FREE_CD3_LYMPH   = -vf_DimerCD3_LY + vb_DimerCD3_LY - vf_Tri_viaCD20_LY +
↪  vb_Tri_toCD20_LY - kdegCD3 * C_FreeCD3_LY * Vlymph + syn_CD3_LY + flow_FreeCD3_LY -
↪  koutTC * C_FreeCD3_LY * Vlymph;
 dxdt_FREE_CD20_LYMPH  = -vf_DimerCD20_LY + vb_DimerCD20_LY - vf_Tri_viaCD3_LY +
↪  vb_Tri_toCD3_LY - kdegCD20 * C_FreeCD20_LY * Vlymph + syn_CD20_LY + flow_FreeCD20_LY
↪  - (koutBC + k_kill_LY) * C_FreeCD20_LY * Vlymph;
```

```
dxdt_DIMER_CD3_LYMPH  = vf_DimerCD3_LY - vb_DimerCD3_LY - vf_Tri_viaCD3_LY +
↪  vb_Tri_toCD3_LY - kintCD3 * C_DimerCD3_LY * Vlymph - koutTC * C_DimerCD3_LY * Vlymph
↪  + flow_DimerCD3_LY + (koutBC + k_kill_LY) * C_Trimer_LY * Vlymph;
dxdt_DIMER_CD20_LYMPH = vf_DimerCD20_LY - vb_DimerCD20_LY - vf_Tri_viaCD20_LY +
↪  vb_Tri_toCD20_LY - kintCD20 * C_DimerCD20_LY * Vlymph + flow_DimerCD20_LY + koutTC *
↪  C_Trimer_LY * Vlymph - (koutBC + k_kill_LY) * C_DimerCD20_LY * Vlymph;
dxdt_TRIMER_LYMPH     = vf_Tri_viaCD3_LY + vf_Tri_viaCD20_LY - vb_Tri_toCD3_LY -
↪  vb_Tri_toCD20_LY + flow_Trimer_LY - (koutTC + koutBC + k_kill_LY) * C_Trimer_LY *
↪  Vlymph;

// Update PK due to Binding
dxdt_PLASMA = dxdt_PLASMA - (vf_DimerCD3_BL - vb_DimerCD3_BL) - (vf_DimerCD20_BL -
↪  vb_DimerCD20_BL);
dxdt_SPLEEN = dxdt_SPLEEN - (vf_DimerCD3_SP - vb_DimerCD3_SP) - (vf_DimerCD20_SP -
↪  vb_DimerCD20_SP);
dxdt_NODE   = dxdt_NODE   - (vf_DimerCD3_LN - vb_DimerCD3_LN) - (vf_DimerCD20_LN -
↪  vb_DimerCD20_LN)
  - (vf_DimerCD20_Tumor - vb_DimerCD20_Tumor);
dxdt_LYMPH  = dxdt_LYMPH  - (vf_DimerCD3_LY - vb_DimerCD3_LY) - (vf_DimerCD20_LY -
↪  vb_DimerCD20_LY);

// ==========================================================
// 8. ACTIVATION SUBMODEL (vATC)
// ==========================================================
// 1. Activation against Tumor (only in node)
double Trimer_per_Tumor = 0.0;
if (Cells_Reachable > 1.0) {
  Trimer_per_Tumor = (CLAMP(TRIMER_TUMOR) / nmol_per_molecule) / Cells_Reachable;
}
double RELU = 0.01;
if (Trimer_per_Tumor > Trimer_Threshold) RELU = 1.0;

double rate_act_Tumor = 0.0;
if (T_SIM > Time_First_Dose + TAD) {
  rate_act_Tumor = RELU * sim_slopetumor * Trimer_per_Tumor;
}

// 2. Activation against B-Cells
double Trimer_per_BC_BL = (BC_BLOOD > 1)  ? (CLAMP(TRIMER_BLOOD) / nmol_per_molecule) /
↪  BC_BLOOD  : 0.0;
double Trimer_per_BC_SP = (BC_SPLEEN > 1) ? (CLAMP(TRIMER_SPLEEN) / nmol_per_molecule)
↪  / BC_SPLEEN : 0.0;
double Trimer_per_BC_LN = (BC_NODE > 1)   ? (CLAMP(TRIMER_NODE) / nmol_per_molecule) /
↪  BC_NODE   : 0.0;
double Trimer_per_BC_LY = (BC_LYMPH > 1)  ? (CLAMP(TRIMER_LYMPH) / nmol_per_molecule) /
↪  BC_LYMPH  : 0.0;

double rate_act_BC_BL = (T_SIM > Time_First_Dose + TAD) ? sim_slope * Trimer_per_BC_BL
↪  : 0.0;
double rate_act_BC_SP = (T_SIM > Time_First_Dose + TAD) ? sim_slope * Trimer_per_BC_SP
↪  : 0.0;
double rate_act_BC_LN = (T_SIM > Time_First_Dose + TAD) ? sim_slope * Trimer_per_BC_LN
↪  : 0.0;
double rate_act_BC_LY = (T_SIM > Time_First_Dose + TAD) ? sim_slope * Trimer_per_BC_LY
↪  : 0.0;
```

```
  // 3. vATC Trafficking
  double flow_vATC_BL_SP = kpt * (1 + INJ_Scaler * INJ) * vATC_BC_BLOOD; // Removed
↪  redundant clamp
  double flow_vATC_SP_LN = ktn * vATC_BC_SPLEEN;
  double flow_vATC_LN_LY = knl * vATC_BC_NODE;
  double flow_vATC_LY_BL = klp * vATC_BC_LYMPH;

  // 4. pATC Trafficking
  double flow_pATC_BL_SP = kpt * (1 + INJ_Scaler * INJ) * pATC_BLOOD;
  double flow_pATC_SP_LN = ktn * pATC_SPLEEN;
  double flow_pATC_LN_LY = knl * pATC_NODE;
  double flow_pATC_LY_BL = klp * pATC_LYMPH;

  // 5. Expansion & Death
  double exp_BL_rate = expand_factor * vATC_BC_BLOOD;
  double exp_SP_rate = expand_factor * vATC_BC_SPLEEN;
  double exp_LN_rate = expand_factor * (vATC_BC_NODE + vATC_TUMOR_NODE);
  double exp_LY_rate = expand_factor * vATC_BC_LYMPH;

  double rate_death_ATC = (T_SIM > (Time_First_Dose + Tp)) ? koutATC : 0.0;

  // 6. ODEs
  dxdt_vATC_BC_BLOOD  = rate_act_BC_BL + flow_vATC_LY_BL - flow_vATC_BL_SP -
↪  rate_death_ATC * vATC_BC_BLOOD;
  dxdt_vATC_BC_SPLEEN = rate_act_BC_SP + flow_vATC_BL_SP - flow_vATC_SP_LN -
↪  rate_death_ATC * vATC_BC_SPLEEN;
  dxdt_vATC_BC_NODE   = rate_act_BC_LN + flow_vATC_SP_LN - flow_vATC_LN_LY -
↪  rate_death_ATC * vATC_BC_NODE;
  dxdt_vATC_BC_LYMPH  = rate_act_BC_LY + flow_vATC_LN_LY - flow_vATC_LY_BL -
↪  rate_death_ATC * vATC_BC_LYMPH;

  dxdt_vATC_TUMOR_NODE = rate_act_Tumor - rate_death_ATC * vATC_TUMOR_NODE;

  dxdt_pATC_BLOOD  = exp_BL_rate + flow_pATC_LY_BL - flow_pATC_BL_SP - rate_death_ATC *
↪  pATC_BLOOD;
  dxdt_pATC_SPLEEN = exp_SP_rate + flow_pATC_BL_SP - flow_pATC_SP_LN - rate_death_ATC *
↪  pATC_SPLEEN;
  dxdt_pATC_NODE   = exp_LN_rate + flow_pATC_SP_LN - flow_pATC_LN_LY - rate_death_ATC *
↪  pATC_NODE;
  dxdt_pATC_LYMPH  = exp_LY_rate + flow_pATC_LN_LY - flow_pATC_LY_BL - rate_death_ATC *
↪  pATC_LYMPH;
}

$TABLE
double Total_pATC = CLAMP(pATC_BLOOD) + CLAMP(pATC_SPLEEN) + CLAMP(pATC_NODE) +
↪  CLAMP(pATC_LYMPH);
double C_Trimer_BL_nM = CLAMP(TRIMER_BLOOD) / Vplasma;
double Tumor_Vol_cm3 = TUMOR_CELLS / 1e9;

$CAPTURE C_plasma Total_pATC C_Trimer_BL_nM Tumor_Vol_cm3
';
writeLines(model_code, "../model/epc_complete_for_rmd.cpp")
```

3. Simulation

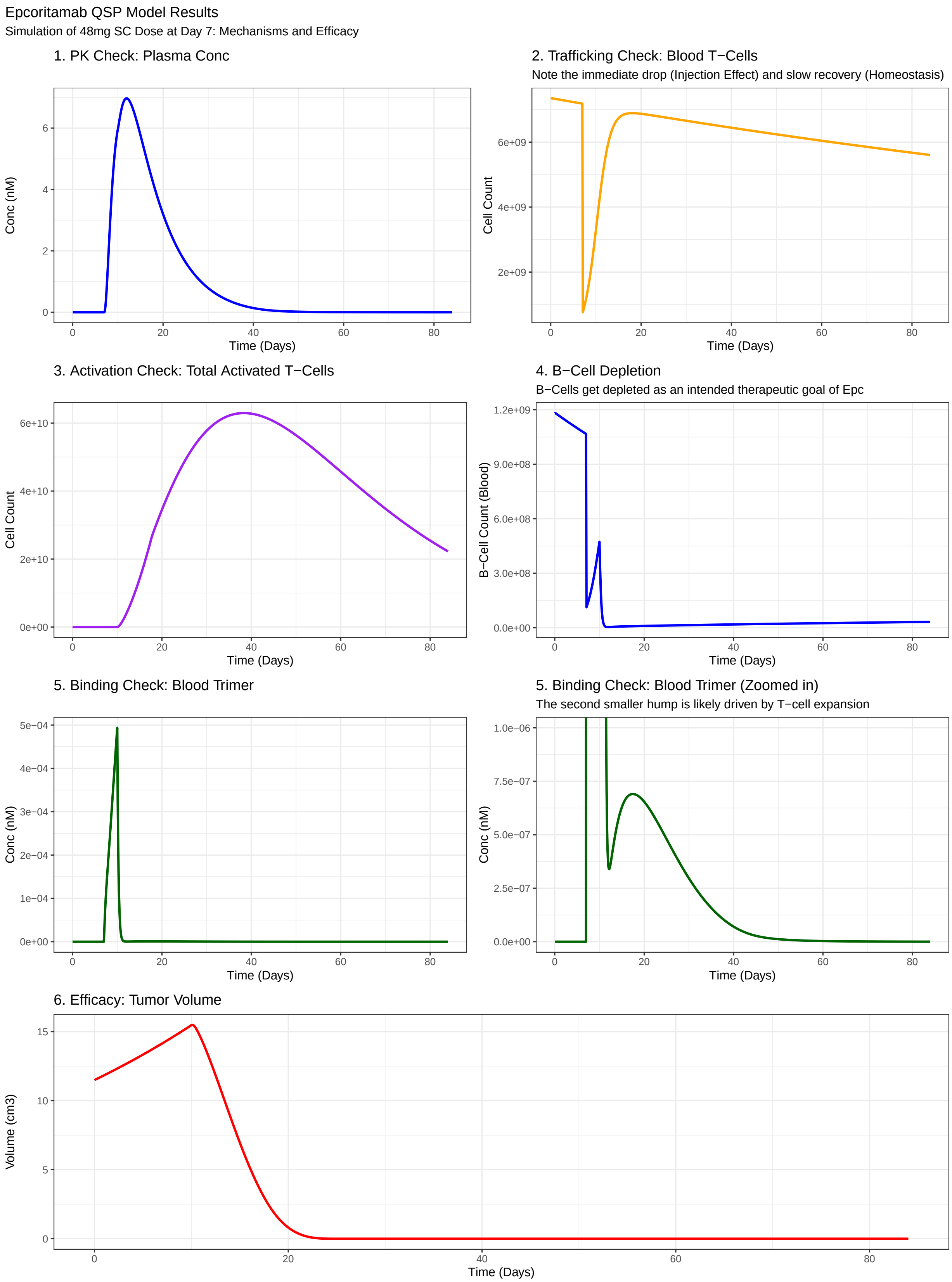
```
# 1. Load the Master Model
mod <- mread("epc_complete_for_rmd", "../model", preclean = TRUE)

# 2. Set up Total Cell Numbers & Define Dosing (48 mg SC)
# 48 mg -> nmol conversion (mg -> g -> mol -> nmol)
dose_nmol <- (48 / 1000 / 145623) * 1e9

# Combined Dose: Drug (DEPOT) + Injection Effect (INJ)
total_dose <- ev(amt = dose_nmol, cmt = "DEPOT", time = 7) +
  ev(amt = 1, cmt = "INJ", time = 7)

V_blood_mm3 <- 5e6
total_TC <- 1469 * V_blood_mm3
total_BC <- 236  * V_blood_mm3

# 3. Run Simulation
# Note: We use a high maxsteps because binding kinetics are fast
sim <- mod %>%
  ev(total_dose) %>%
  param(Time_First_Dose = 7, scale_binding = 1, kkill_tumor = 0.165e-6) %>%
  init(
    TC_BLOOD = total_TC,
    BC_BLOOD = total_BC,
    TC_SPLEEN = total_TC * 30,
    BC_SPLEEN = total_BC * 30,
    TC_NODE = total_TC * 0.03,
    BC_NODE = total_BC * 0.03,
    TC_LYMPH = total_TC * 19,
    BC_LYMPH = total_BC * 19,
    AF_TC = 1,
    AF_BC = 1,
    INJ = 0
  ) %>%
  # Run for 12 weeks (84 days)
  mrgsim(end = 84, delta = 0.1,
         atol = 1e-3, rtol = 1e-3,
         maxsteps = 1e9) %>%
  as_tibble()
print('Finish running the simulation')
```

```
## [1] "Finish running the simulation"
```

```
# --- 4. DEBUGGING PLOTS (The Chain of Causality) ---

# A. PK Check: Is drug getting into the system?
p1 <- ggplot(sim, aes(x = time, y = C_plasma)) +
  geom_line(color = "blue", linewidth = 1) +
  labs(title = "1. PK Check: Plasma Conc", y = "Conc (nM)", x = "Time (Days)") +
  theme_bw()

# B. Trafficking Check: Are T-Cells leaving the blood?
```

```r
p2 <- ggplot(sim, aes(x = time, y = TC_BLOOD)) +
  geom_line(color = "orange", linewidth = 1) +
  labs(
    title = "2. Trafficking Check: Blood T-Cells",
    subtitle = "Note the immediate drop (Injection Effect) and slow recovery
    ↪  (Homeostasis)",
    y = "Cell Count",
    x = "Time (Days)"
  ) +
  theme_bw()

# C. Binding Check: Are Trimers forming in the blood?
p3 <- ggplot(sim, aes(x = time, y = C_Trimer_BL_nM)) +
  geom_line(color = "darkgreen", linewidth = 1) +
  labs(title = "5. Binding Check: Blood Trimer", y = "Conc (nM)", x = "Time (Days)") +
  theme_bw()

p3_zoomed <- ggplot(sim, aes(x = time, y = C_Trimer_BL_nM)) +
  geom_line(color = "darkgreen", linewidth = 1) +
  labs(title = "5. Binding Check: Blood Trimer (Zoomed in)",
       subtitle = "The second smaller hump is likely driven by T-cell expansion",
       y = "Conc (nM)", x = "Time (Days)") +
  coord_cartesian(ylim = c(0, 1e-6)) +
  theme_bw()

p_bcell <- ggplot(sim, aes(x = time, y = BC_BLOOD)) +
  geom_line(color = "blue", linewidth = 1) +
  labs(
    title = "4. B-Cell Depletion",
    subtitle = "B-Cells get depleted as an intended therapeutic goal of Epc",
    y = "B-Cell Count (Blood)",
    x = "Time (Days)"
  ) +
  scale_y_continuous(labels = scales::scientific) +
  # coord_cartesian(xlim = c(0, 10)) + # Zoom in on the crash
  theme_bw()


# D. Activation Check: Are T-cells expanding?
p4 <- ggplot(sim, aes(x = time, y = Total_pATC)) +
  geom_line(color = "purple", linewidth = 1) +
  labs(title = "3. Activation Check: Total Activated T-Cells", y = "Cell Count", x =
  ↪  "Time (Days)") +
  theme_bw()

# E. Efficacy Check: Is the Tumor shrinking?
p5 <- ggplot(sim, aes(x = time, y = Tumor_Vol_cm3)) +
  geom_line(color = "red", linewidth = 1) +
  labs(title = "6. Efficacy: Tumor Volume", y = "Volume (cm3)", x = "Time (Days)") +
  theme_bw()

library(patchwork)

# Layout:
```

```r
# Row 1: PK | Trafficking
# Row 2: Activation | B-Cells
# Row 3: Trimer | Trimer Zoomed
# Row 4: Tumor (Full Width)
layout <- (p1 + p2) / (p4 + p_bcell) / (p3 + p3_zoomed) / p5

# Add Main Title and Print
layout + plot_annotation(
  title = 'Epcoritamab QSP Model Results',
  subtitle = 'Simulation of 48mg SC Dose at Day 7: Mechanisms and Efficacy'
)
```


Epcoritamab QSP Model Results
Simulation of 48mg SC Dose at Day 7: Mechanisms and Efficacy
1. PK Check: Plasma Conc
Conc (nM)
0
2
4
6
0
20
40
60
80
Time (Days)
2. Trafficking Check: Blood T−Cells
Note the immediate drop (Injection Effect) and slow recovery (Homeostasis)
Cell Count
2e+09
4e+09
6e+09
0
20
40
60
80
Time (Days)
3. Activation Check: Total Activated T−Cells
Cell Count
0e+00
2e+10
4e+10
6e+10
0
20
40
60
80
Time (Days)
4. B−Cell Depletion
B−Cells get depleted as an intended therapeutic goal of Epc
B−Cell Count (Blood)
0.0e+00
3.0e+08
6.0e+08
9.0e+08
1.2e+09
0
20
40
60
80
Time (Days)
5. Binding Check: Blood Trimer
Conc (nM)
0e+00
1e−04
2e−04
3e−04
4e−04
5e−04
0
20
40
60
80
Time (Days)
5. Binding Check: Blood Trimer (Zoomed in)
The second smaller hump is likely driven by T−cell expansion
Conc (nM)
0.0e+00
2.5e−07
5.0e−07
7.5e−07
1.0e−06
0
20
40
60
80
Time (Days)
6. Efficacy: Tumor Volume
Volume (cm3)
0
5
10
15
0
20
40
60
80
Time (Days)

4. Insights & Discussion

The simulation of the Epcoritamab mPBPK-QSP model yielded several key insights into the pharmacology of T-cell engaging bispecific antibodies. These can be categorized into biological mechanisms and technical modeling challenges.

A. Biological & Clinical Insights 1. The "Kamikaze" Pharmacodynamics of Target Depletion A unique feature of this bispecific antibody is that its pharmacological activity (killing) destroys the very scaffold required for its activity (the target B-cell).
* **Insight:** The Trimer concentration profile (Panel 5) is **biphasic**.
* *Phase 1 (The Spike):* Rapid binding to the large baseline B-cell population.
* *Phase 2 (The Crash):* As T-cells activate and kill B-cells, the "anchors" for the trimers are destroyed, causing trimer levels to plummet even while drug levels are high.
* *Phase 3 (The Hump):* A secondary wave of trimer formation (Panel 5 zoomed in) occurs as the T-cell army expands ($pATC$), increasing the availability of CD3 binding sites.
* **Clinical Implication:** Efficacy is self-limiting in a way that traditional chemotherapy is not. The drug needs to recruit T-cells fast enough to kill the tumor *before* it runs out of B-cell targets to anchor the synapse.

2. B-Cell Aplasia is a Feature, Not a Bug The model successfully reproduces profound B-cell depletion (Panel 4).
* **Insight:** Even as drug levels decline (Panel 1), B-cell counts remain suppressed near zero. This occurs because the killing rate (driven by drug levels far above the K_D) vastly exceeds the bone marrow's natural production rate ($\approx 2 \times 10^8$ cells/day).
* **Clinical Implication:** This confirms the mechanism of prolonged immunosuppression observed clinically, necessitating immunoglobulin replacement therapy in some patients. Recovery only begins when drug levels drop significantly below the effective concentration.

3. The Injection Effect (Margination)
The model captures the transient drop in peripheral lymphocytes immediately after dosing (Panel 2).
* **Insight:** This is not due to cell death, but rather **margination**—the trafficking of cells from the blood into tissues (Spleen) triggered by the injection event. The homeostatic feedback loop ($AF_{TC/BC}$) correctly identifies this as a "loss" and upregulates production, leading to the subsequent rebound.

B. Technical & Modeling Insights 1. Handling Stiff Systems in QSP
The model presents a classic "stiff" differential equation problem: binding reactions occur on the scale of seconds/minutes (k_{on}), while cell growth occurs over days/weeks.
* **Challenge:** Standard solvers failed with `maxsteps` errors or required infinitesimal step sizes.
* **Solution:** I successfully stabilized the model by using a robust ODE solver configuration (`atol/rtol = 1e-3`) and verifying that parameter scaling (e.g., `scale_binding`) did not alter the equilibrium state. This highlights the trade-off between physical precision (microseconds) and clinical relevance (weeks).

2. Mass Conservation in Receptor Recycling A critical implementation detail was ensuring mass conservation during cell death.
* **Insight:** When a cell dies (naturally or via killing), the surface complexes (Dimers and Trimers) do not simply vanish.
* **Implementation:** I modeled the "Recycling" pathway: when a B-cell dies, the $Drug - CD3 - CD20$ trimer breaks, releasing the $Drug - CD3$ dimer back into the T-cell pool. Omitting this term would lead to an artificial "sink" of the drug and underestimate T-cell engagement.

3. Unit Scaling is Critical for "Relative" Parameters
Parameters from literature (e.g., `sim_slope`, `kkill`) are sometimes reported in "per million cells" units, whereas ODE models track single cells.
* **Insight:** Direct implementation of literature values led to "instant wipeout" events (killing rates 10^{12} times too fast).

* **Correction:** Explicit dimensional analysis and scaling (multiplying/dividing by 10^6) were required to translate population-level parameters into cell-level mechanistic logic.